QI WU

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EDUCATION

University of Science and Technology of China — Hefei, China
Bachelor of Data Science — Sep 2023 - present
GPA: 3.64/4.3(87/100)

University of California, Berkeley — Berkeley, CA, USA
Exchange Undergraduate Student — Jan. 2026 - May. 2026

RESEARCH INTERESTS

Trustworthy Artificial Intelligence, Multimodal Large Language Models, Medical Vision-Language Models, Uncertainty Quantification, LLM Alignment and Preference Optimization;

PUBLICATION AND UNDER REVIEW MANUSCRIPTS

[1] Q. Wu†, X. Liu†, A. Wang, R. Wang, W. Chen, W. Wang, and Q. Yao. "Rethinking Closed-Ended Medical VLM Evaluation: From Template Shortcuts to Counterfactual Answer Sensitivity." *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 2026. Accepted.

[2] X. Liu†, Q. Wu†, A. Wang, M. Li, W. Chen, W. Wang, R. Wang, H. Lai, S. K. Zhou, and Q. Yao. "Clinical Decision-Token Evidence Discrepancy for Medical VQA Uncertainty Quantification." *Submitted to IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2027. Under Review.

† Equal contribution.

RESEARCH EXPERIENCE

Trustworthy Medical Vision-Language Models — USTC

Advisor: **Prof.** Qingsong Yao — Jan.2026 – Present

Counterfactual Answer Sensitivity for Reliable Medical VLM Evaluation

• Identified stable wrongness in closed-ended medical VLMs, where models produce confident but incorrect predictions despite corrupted or missing visual evidence.
• Proposed Counterfactual Answer Sensitivity (CAS), a vision-grounded uncertainty framework using counterfactual visual an interventions d evidence sensitivity features for reliability assessment.
• Evaluated across 7 VLMs and 3 VQA benchmarks (VQA-RAD, SLAKE, VQA-Med-2019), achieving +0.150 AUROC improvement over HAC-based uncertainty methods.

ClinEviD: Decision-Token Evidence Discrepancy for Medical VQA Uncertainty Quantification

• Identified decision-token dilution in medical VQA uncertainty estimation, where response-level aggregation overlooks critical clinical tokens.
• Developed ClinEviD, a training-free, single-pass framework that localizes evidential uncertainty to decision-bearing tokens through token-level evidence discrepancy modeling.
• Designed token-level evidence discrepancy modeling by contrasting generated clinical tokens with their strongest competing alternatives, followed by top-k aggregation and lightweight logistic calibration for answer-level risk estimation.
• Evaluated on 7 VLMs, 3 medical VQA datasets, and 42 model–dataset settings, achieving 0.7535 mean AUROC, +0.0798 AUROC over HAC, and 18× lower inference cost.

The Exploration of Preference Optimization in LLMs for Recommendation Systems — USTC

Advisor: **Prof.** Xiangnan He — January. 2025 – June.2025

• Explored and optimized existing alignment (RLHF) and preference optimization methods, such as PPO, DPO, and SimPO.
• Explored LLMs' preference optimization capabilities to improve the performance of recommendation systems.
• Explore to promote negative sampling in LLM for recommendation system.

HONORS & AWARDS

• Outstanding Student Scholarship, Silver award (top 10%) — 2024
• The Chinese Mathematics Competitions for College Students(CMC), Second Prize — 2024

SKILLS

Programming: C, Python (Pytorch), LATEX
English: CET-4 (622), Tofel (5.5), GRE (316)